

Group 3 Project Proposal + Annotated Bibliography

Group Members:

Farid Mohammed, Daikun Zhu, Ke Zhang, Yizuo Wu, Zicheng Liu, Archy Hwang

Research Description:

Our research focuses on the global spread and impact of misinformation and disinformation, exploring how false information influences public understanding, health behavior, and trust. Drawing on interdisciplinary sources, we examine both theoretical and real-world perspectives. Denniss and Lindberg (2025) analyze how misinformation spreads like an infectious disease on social media, while Broda and Strömbäck (2024) provide a comprehensive framework distinguishing misinformation, disinformation, and fake news. Additional studies, such as Lewandowsky et al. (2012) and Vosoughi et al. (2018), explain the psychological and algorithmic mechanisms that make false information more persistent and viral. Reports from Pew Research Center (2019) and the Council of Europe (Wardle & Derakhshan, 2017) highlight public concern and institutional responses. Together, these sources form the foundation of our project, which aims to visualize how misinformation circulates across networks and to analyze its social consequences. This research will guide our proposal's design and ensure that our final presentation is grounded in credible evidence and critical information literacy principles.

Proposal:

Using a website builder, we will build an interactive flowchart displaying how information is altered (or in some cases, completely fabricated) to become misinformation or disinformation, as well exploring the public reactions to both.

This flowchart will have several “stages”/“nodes”, where the user can click and explore each distinct pathway of the alteration/fabrication of information.

With each “stage”, there'd be a small description and an example—and if needed/helpful, a small drawn representation—of a given stage.

Here's a small example of possible flowchart logic:

Fact → Mishearing → Propagation

>> User clicks on “Fact”; Fact: The simplest block of information; *it is an observation or basic true statement about something.*

>> User clicks on Mishearing; Mishearing: Someone hears part or “bits and pieces” of a fact, *getting a skewed, misleading, or partial understanding of the fact.*

>> User clicks on Propagation; Propagation: This misheard fact is then propagated in certain circles, *leading to a misinformed view.*

To finish the website in a timely manner:

Week 1: Drafting and Finishing the Misinformation/Disinformation Flowchart.

Week 2 Detailing “Stages”/“Nodes” with descriptions, examples, drawn representations. Linking sub-pages for easier browsing can be done here as well.

Week 3: Polishing “Stages”/“Nodes” to make sure they are functional, readable, relevant, and informative.

To distribute the workload accordingly:

Due to the small, compressed scope of our project (a single webpage with multiple sub-pages of information accessed through UI on the main webpage), every group member can contribute to any sub-page, although it’s more likely that each group member will work on their own section of the flowchart, when created.

Annotated Bibliography:

Source 1:

Vosoughi Soroush, Sinan Aral, and Deb Roy.

“The Spread of True and False News Online.” *Science*, vol. 359, no. 6380, 2018, pp. 1146–1151.
<https://doi.org/10.1126/science.aap9559>

Annotation:

This article studies how false news spreads on social media, especially on Twitter. The authors found that false information spreads faster, further, and more widely than true information. They explain that this happens because false news is often more novel and emotionally engaging.

This source is highly credible because it is published in *Science*, one of the most respected academic journals. It is useful for my project because it provides strong evidence that disinformation spreads more effectively than accurate information on social media platforms.

Source 2:

Claire Wardle and Hossein Derakhshan.

“Information Disorder: Toward an Interdisciplinary Framework.” Council of Europe, 2017.
<https://rm.coe.int/information-disorder-report/168076277c>

Annotation:

This report introduces a framework for understanding misinformation, disinformation, and malinformation. The authors explain how different types of false or misleading information are created and spread. They also discuss the roles of technology, media, and users in the spread of information disorder.

This source is credible because it is produced by the Council of Europe and widely cited in research on misinformation. It is useful for my project because it provides a clear framework for analyzing different types of false information.

Source 3:

Ferreira Caceres, M. M., Sosa, J. P., Lawrence, J. A., Sestacovschi, C., Tidd-Johnson, A., Rasool, M. H. U., Gadamidi, V. K., Ozair, S., Pandav, K., Cuevas-Lou, C., Parrish, M., Rodriguez, I., & Fernandez, J. P. (2022, January 12). *The impact of misinformation on the COVID-19 pandemic*. AIMS public health.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9114791/>

Annotation:

This article talks about how misinformations affected the COVID-19 pandemic, especially on social media. It shows the speed of false information being spread online, which makes things worse. For example, some people refused to trust vaccines or refused to wear masks because of comments they saw on social media. The article explains how this kind of information can confuse people and lead to bad decisions.

I think this source is reliable because it is a research article based on real data, not just opinions from others. It is useful for our project because it gives a clear real-life example of how disinformation can actually impact people during important events like a pandemic.

Source 4:

Mitchell, A., Gottfried, J., Stocking, G., Walker, M., & Fedeli, S. (2019, June 5). *Many Americans say made-up news is a critical problem that needs to be fixed*. Pew Research Center.
<https://www.pewresearch.org/journalism/2019/06/05/many-americans-say-made-up-news-is-a-critical-problem-that-needs-to-be-fixed/>

Annotation:

This article talks about different views from people in the US about misinformation and how serious they think it is. It shows that a lot of residents believe that fake news is a big problem and it can affect how people understand serious situations. The article also explains that fake news can create confusion and make it harder for people to trust information.

I feel like this source is reliable because it comes from Pew Research Center. It is known for real surveys and data. This source is useful for our project because it shows how misleading news affects people's views and why it matters in real life.

Source 5:

Denniss, Emily, and Rebecca Lindberg. "Social Media and the Spread of Misinformation: Infectious and a Threat to Public Health." *Health Promotion International*, vol. 40, no. 2, 31 Mar. 2025, pp. 1–10, academic.oup.com/heapro/article/40/2/daaf023/8100645?searchresult=1, <https://doi.org/10.1093/heapro/daaf023>.

Annotation:

This article examines how misinformation spreads across social media platforms and why it poses a significant threat to public health. Denniss and Lindberg argue that misinformation behaves similarly to infectious disease: it spreads rapidly, exploits emotional triggers, and disproportionately affects vulnerable populations. The authors analyze case studies from recent health crises, showing how false claims about treatments, vaccines, and disease origins can undermine trust in institutions and lead to harmful real-world behaviors. Published in *Health Promotion International*, a peer-reviewed journal from Oxford University Press, the article is highly credible, supported by empirical evidence and grounded in public health research. This source is valuable for our project because it provides a clear explanation of the mechanisms behind misinformation spread and its societal consequences. In our proposal, we can use this article to support the section discussing the real-world impact of misinformation, especially when visualizing how false information circulates through social networks and affects public decision-making.

Source 6:

Broda, Elena, and Jesper Strömbäck. "Misinformation, Disinformation, and Fake News: Lessons from an Interdisciplinary, Systematic Literature Review." *Annals of the International Communication Association*, vol. 48, no. 2, 6 Mar. 2024, pp. 139–166,

www.tandfonline.com/doi/full/10.1080/23808985.2024.2323736,
<https://doi.org/10.1080/23808985.2024.2323736>.

Annotation:

Broda and Strömbäck present an extensive interdisciplinary systematic literature review that synthesizes research from communication studies, psychology, political science, and information science. The article clarifies the conceptual differences between misinformation, disinformation, and fake news, emphasizing how each term reflects distinct intentions, mechanisms, and societal impacts. The authors identify recurring themes across the literature, including cognitive biases that make individuals vulnerable to false information, the role of social media algorithms in amplifying misleading content, and the broader decline in institutional trust that enables misinformation to spread more easily. Because the review draws on findings from dozens of peer-reviewed studies, it offers a comprehensive theoretical foundation and highlights gaps in current scholarship, such as the need for more cross-cultural research and better measurement tools. Both authors are respected scholars in political communication, and the journal is a reputable, peer-reviewed academic publication, which enhances the credibility of the source. This article will support our proposal by helping us define key concepts, establish a strong theoretical framework, and justify why misinformation and disinformation are critical issues that require deeper analysis and visualization.

Source 7:

Lewandowsky, Stephan, et al. "Misinformation and Its Correction: Continued Influence and Successful Debiasing." *Psychological Science in the Public Interest*, vol. 13, no. 3, 2012, pp. 106–131, <https://doi.org/10.1177/1529100612451018>.

Annotation:

This article explores why misinformation and disinformation continue to influence people even after they have been corrected. Lewandowsky and his co-authors argue that once false information is accepted, it becomes difficult to remove from memory due to cognitive biases and mental shortcuts. The study also discusses strategies for reducing the effects of disinformation, such as providing clear and repeated corrections. Published in a peer-reviewed psychology journal, the article is highly credible and grounded in scientific research. This source is useful for our project because it explains the psychological reasons why disinformation is so persistent and difficult to counter, which is important for understanding its real-world impact.

Source 8:

Benkler, Yochai, Robert Faris, and Hal Roberts. *Network Propaganda: Manipulation*,

Disinformation, and Radicalization in American Politics. Oxford University Press, 2018,
<https://doi.org/10.1093/oso/9780190923624.001.0001>.

Annotation:

This book analyzes how disinformation spreads within political media systems, particularly during the 2016 U.S. presidential election. Benkler and his co-authors argue that disinformation is reinforced by partisan media networks, not just social media platforms. They show how certain media ecosystems amplify misleading narratives and contribute to political polarization.

Published by Oxford University Press, this work is highly credible and based on extensive data analysis. This source is valuable for our project because it explains how disinformation operates at a systemic level, showing the role of media structures in shaping public perception.